

Modulus equations and inequalities

Key Points

Modulus $|f(x)| = \begin{cases} f(x) & \text{if } f(x) \geq 0 \\ -f(x) & \text{if } f(x) < 0 \end{cases}$

To solve $|f(x)| = g(x)$ solve both $\pm f(x) = g(x)$. Another way is to square to remove the modulus as

$$|f(x)|^2 = (f(x))^2$$

However limits on the domain of x for $f(x) \geq 0$ and $f(x) < 0$ limit solutions

Similarly inequalities are solved with these limits on the domain for both + and - versions

Basic Skills

Q1. Solve the following equations

a) $2x - 4 = 6$

b) $4 - 2x = 6$

c) Hence state the solutions to $|2x - 4| = 6$

Q2. Solve the equation $|6x + 2| = 10$

Q3. Solve the following inequalities

a) $5x - 4 > 6$

b) $4 - 5x > 6$

Q4. Solve $|5 - 2x| = x$

Competency Skills

Q5. a) Solve the equation $|3x + 2| = 5$

b) State the domain for x such that $3x + 2 \geq 0$ and then such that $3x + 2 < 0$

c) Solve both $3x + 2 < 5$ and $-3x - 2 < 5$ and hence show the solutions to $|3x + 2| > 5$ are $-\frac{2}{3} < x < 1$ and $-\frac{7}{3} < x < -\frac{2}{3}$

d) Solve $|3x + 2| > 5$

Q6. Solve the inequality $|6x - 4| > 2x$

Q7. Solve $|\tan x| = 1$ for $-\pi < x \leq \pi$

Q8. Solve $|2x - 1| = 2|x| - 1$

Q9. Solve $|e^{4x-2}| \geq |-2|$

Advanced Skills

Q10. a) Solve $|\cos x| = \frac{1}{2}$ for $-\pi < x \leq \pi$

b) Hence solve $|\cos x| \leq \frac{1}{2}$ for $-\pi < x \leq \pi$

Q11. Solve $|2 - 4x| = 4|x + 2| - 1$

Q12. Solve the inequality $|3x - 2| + 2x > 5 - x$

Q13. How many real solutions can the equation $|x^2 + b| = c$ have. Explain the cases depending on b, c

Q14. Solve $||x|^2 - 4|x|| = 2$

Extension

Q15. What happens when considering the transformation of a graph $y = f(x)$ to $y = f(|x|)$

Hint: Try using a graph to see what happens when taking the magnitude of the input

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